Dachuan Song

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Google Scholar | GitHub | LinkedIn | Personal Website

Education

George Mason University, Fairfax, Virginia, USA — Jan 2024 – Present
Ph.D. in Electrical and Computer Engineering; GPA: 3.91/4.00 — *Expected: December 2027*
Advisor: Prof. Xuan Wang

University of Southampton, Southampton, United Kingdom — Sept 2022 – Sept 2023
M.Sc. in Computer Science; GPA: 3.8 (Top 3%)

Xinjiang University, Xinjiang, China — Sept 2017 – Jun 2021
B.Eng. in Software Engineering; GPA: 3.4 (Top 5%)

Selected Research Projects

Elastic Spectral State Space Models (ES-SSM) — Jan 2026 – May 2026
First author, ***NeurIPS 2026*** *Main Track (Poster).* Code: github.com/songdc98/ES-SSM

- Developed a train-once spectral state-space model that exports compact models for different compute budgets by direct prefix truncation, without budget-specific retraining.
- Combined input-adaptive channel-wise gates with budget dropout to optimize the same spectral prefixes used at inference. Evaluated quality–cost trade-offs across language modeling, long-range sequence tasks, audio classification, and offline reinforcement learning.

Elastic Selective Spectral Hybrids (ESSH) — Jul 2026 – Sep 2026

- Developed selective spectral recurrent channels with input-dependent decay and write/read gates, combined with sliding-window attention for language modeling.
- Jointly trained spectral-channel count and feed-forward width to produce compact model exports from one checkpoint, avoiding computation for discarded channels.
- Benchmarked **1.53B-parameter** fused decoding on an NVIDIA A100 with matched parameter counts and the same compilation policy for all models. ESSH matched Mamba-2 in speed, with **1.1–1.6**$\times$ speedups over Mamba-3 and **1.1–4.6**$\times$ over Transformer++.

State-Space Language Models — Sep 2026 – Present

- Extending state-space language models to support **exact recall and multi-hop reasoning** over long contexts.

Internship Experience

Machine Learning Engineer Intern, Optosurgical — May 2026 – Aug 2026
Columbia, Maryland, USA

- Developed video models to recognize surgical phases and instrument actions, providing procedure-state signals to downstream systems.
- Compressed and fine-tuned Depth Anything 3 to **29% of its original model size** while retaining near-baseline depth-estimation performance.

Publications

Conference Papers and Preprints:

Elastic Spectral State Space Models for Train-Once Budgeted Inference 2026

Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang

https://arxiv.org/abs/2601.22488

*Accepted to **NeurIPS 2026** Main Track (Poster); proceedings forthcoming. arXiv:2601.22488, 2026.*

Elastic Selective Spectral Hybrids for Train-Once, Export-Many Budgeted Inference Sep 2026

Dachuan Song, Chuchu Chen, Xuan Wang

https://arxiv.org/abs/2609.32486

Mergeable Model-Side Aggregation States for Long-Context Language Models Jul 2026

Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang

https://arxiv.org/abs/2607.26448

Causality-based Subject and Task Fingerprints using fMRI Time-series Data Nov 2024

Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang

https://doi.org/10.1145/3698587.3701342

*ACM BCB '24, Article 13, pp. 1–10 (**Oral Presentation**).*

Journal Articles:

Reconstructing brain causal dynamics for subject and task fingerprints using fMRI time-series data Oct 2025

Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang

https://doi.org/10.1007/s13755-025-00388-w

Health Information Science and Systems 13, Article 70 (2025).

Technical Skills

- **Programming & ML Systems:** Python, PyTorch, CUDA, multi-GPU training, FFT convolution, fused recurrent decoding, reproducible benchmarking.
- **Models & Inference:** large language models (LLMs), recurrent memory, sliding-window attention, model compression, multi-budget model export.
- **State-Space Models:** spectral and selective SSMs; Mamba, Mamba-2, and Mamba-3.
- **Delta-Rule Linear Attention:** Gated DeltaNet, Kimi Delta Attention (KDA), and Gated DeltaNet-2.
- **Evaluation:** language modeling, long-context memory, retrieval and reasoning, controlled ablations, latency, throughput, and memory profiling.